

# full\_robustness\_sensitivity\_results

| Family              | Gate  | Scenario                    | Specification | N   | Term         | Metric | Estimate    | CI_Low     | CI_High    | p_value     | Significant | Direction | Formatted              |
|---------------------|-------|-----------------------------|---------------|-----|--------------|--------|-------------|------------|------------|-------------|-------------|-----------|------------------------|
| Repository-level    | Gate1 | Repository-clustered        | C/SE + PR     | 312 | Context      | OR     | 3.08728     | 1.6416     | 5.80609    | 0.000468641 | True        | positive  | 3.09*** [1.64, 5.81]   |
| Repository-level    | Gate1 | Repository-clustered        | C/SE + PR     | 312 | Specificity  | OR     | 5.91753     | 3.20686    | 10.9194    | 1.28481e-08 | True        | positive  | 5.92*** [3.21, 10.92]  |
| Repository-level    | Gate1 | Repository-clustered        | C/SE + PR     | 312 | Verification | OR     | 0.947243    | 0.491697   | 1.82484    | 0.871298    | False       | negative  | 0.95 [0.49, 1.82]      |
| Repository-level    | Gate1 | Repository-clustered        | C/SE + PR     | 312 | Log_PR_Size  |        | 1.03208     | 0.861847   | 1.23593    | 0.731365    | False       | positive  | 1.03 [0.86, 1.24]      |
| Repository-level    | Gate1 | Repository-clustered        | C/SE + PR     | 182 | Context      | OR     | 2.0098      | 0.895715   | 4.5096     | 0.0904807   | False       | positive  | 2.01 [0.90, 4.51]      |
| Repository-level    | Gate1 | Repository-clustered        | C/SE + PR     | 182 | Specificity  | OR     | 6.82054     | 2.17349    | 21.4033    | 0.00100017  | True        | positive  | 6.82** [2.17, 21.40]   |
| Repository-level    | Gate1 | Repository-clustered        | C/SE + PR     | 182 | Verification | OR     | 3.61642     | 1.49083    | 8.77262    | 0.00446628  | True        | positive  | 3.62** [1.49, 8.77]    |
| Repository-level    | Gate1 | Repository-clustered        | C/SE + PR     | 182 | Log_PR_Size  |        | 1.47276     | 1.15915    | 1.87121    | 0.00153038  | True        | positive  | 1.47** [1.16, 1.87]    |
| Repository-level    | Gate2 | Repository-clustered        | C/SE + PR     | 872 | Context      | AME    | 0.137641    | 0.0289086  | 0.246374   | 0.0130993   | True        | positive  | 0.138* [0.029, 0.24]   |
| Repository-level    | Gate2 | Repository-clustered        | C/SE + PR     | 872 | Specificity  | AME    | -0.0286261  | 0.152462   | 0.0952102  | 0.6505      | False       | negative  | -0.029 [-0.152, 0.095] |
| Repository-level    | Gate2 | Repository-clustered        | C/SE + PR     | 872 | Verification | AME    | 0.000620507 | 0.102046   | 0.103287   | 0.990548    | False       | positive  | 0.001 [-0.102, 0.103]  |
| Repository-level    | Gate2 | Repository-clustered        | C/SE + PR     | 872 | Log_PR_Size  | AME    | -0.0373924  | 0.0778061  | 0.00302153 | 0.697658    | False       | negative  | -0.037 [-0.078, 0.003] |
| Dominant repository | Gate1 | Stirling largest repository | C/SE + PR     | 302 | Context      | OR     | 2.79318     | 1.53801    | 5.0727     | 0.000740836 | True        | positive  | 2.79*** [1.54, 5.07]   |
| Dominant repository | Gate1 | Stirling largest repository | C/SE + PR     | 302 | Specificity  | OR     | 5.77609     | 2.74513    | 12.1536    | 3.82663e-06 | True        | positive  | 5.78*** [2.75, 12.15]  |
| Dominant repository | Gate1 | Stirling largest repository | C/SE + PR     | 302 | Verification | OR     | 0.928022    | 0.460679   | 1.86947    | 0.83441     | False       | negative  | 0.93 [0.46, 1.87]      |
| Dominant repository | Gate1 | Stirling largest repository | C/SE + PR     | 302 | Log_PR_Size  |        | 1.02284     | 0.854318   | 1.22461    | 0.805778    | False       | positive  | 1.02 [0.85, 1.22]      |
| Dominant repository | Gate1 | Stirling top 5 repositories | C/SE + PR     | 182 | Context      | OR     | 2.80939     | 1.50536    | 5.24304    | 0.00117507  | True        | positive  | 2.81** [1.51, 5.24]    |
| Dominant repository | Gate1 | Stirling top 5 repositories | C/SE + PR     | 182 | Specificity  | OR     | 5.70206     | 2.53525    | 12.8246    | 2.55927e-05 | True        | positive  | 5.70*** [2.54, 12.82]  |
| Dominant repository | Gate1 | Stirling top 5 repositories | C/SE + PR     | 182 | Verification | OR     | 0.976033    | 0.459983   | 2.07104    | 0.949607    | False       | negative  | 0.98 [0.46, 2.07]      |
| Dominant repository | Gate1 | Stirling top 5 repositories | C/SE + PR     | 182 | Log_PR_Size  |        | 1.03419     | 0.854903   | 1.25109    | 0.729251    | False       | positive  | 1.03 [0.85, 1.25]      |
| Dominant repository | Gate1 | Stirling largest repository | C/SE + PR     | 182 | Context      | OR     | 1.97621     | 0.823042   | 4.74511    | 0.127459    | False       | positive  | 1.98 [0.82, 4.75]      |
| Dominant repository | Gate1 | Stirling largest repository | C/SE + PR     | 182 | Specificity  | OR     | 6.48325     | 2.04293    | 20.5747    | 0.00151186  | True        | positive  | 6.48** [2.04, 20.57]   |
| Dominant repository | Gate1 | Stirling largest repository | C/SE + PR     | 182 | Verification | OR     | 3.73407     | 1.47155    | 9.47524    | 0.00555272  | True        | positive  | 3.73** [1.47, 9.48]    |
| Dominant repository | Gate1 | Stirling largest repository | C/SE + PR     | 182 | Log_PR_Size  |        | 1.43926     | 1.13232    | 1.8294     | 0.00292592  | True        | positive  | 1.44** [1.13, 1.83]    |
| Dominant repository | Gate1 | Stirling top 5 repositories | C/SE + PR     | 112 | Context      | OR     | 1.96531     | 0.791311   | 4.88109    | 0.145482    | False       | positive  | 1.97 [0.79, 4.88]      |
| Dominant repository | Gate1 | Stirling top 5 repositories | C/SE + PR     | 112 | Specificity  | OR     | 5.4365      | 1.64586    | 17.9575    | 0.00548173  | True        | positive  | 5.44** [1.65, 17.96]   |
| Dominant repository | Gate1 | Stirling top 5 repositories | C/SE + PR     | 112 | Verification | OR     | 3.48916     | 1.2954     | 9.39806    | 0.0134388   | True        | positive  | 3.49* [1.30, 9.40]     |
| Dominant repository | Gate1 | Stirling top 5 repositories | C/SE + PR     | 112 | Log_PR_Size  |        | 1.38946     | 1.08407    | 1.78087    | 0.00939268  | True        | positive  | 1.39** [1.08, 1.78]    |
| Dominant repository | Gate2 | Stirling largest repository | C/SE + PR     | 872 | Context      | AME    | 0.143157    | -0.0972178 | 0.383532   | 0.243101    | False       | positive  | 0.143 [-0.097, 0.38]   |
| Dominant repository | Gate2 | Stirling largest repository | C/SE + PR     | 872 | Specificity  | AME    | -0.0170431  | 0.262528   | 0.228442   | 0.891764    | False       | negative  | -0.017 [-0.263, 0.228] |
| Dominant repository | Gate2 | Stirling largest repository | C/SE + PR     | 872 | Verification | AME    | -0.00654184 | 0.222942   | 0.216336   | 0.954124    | False       | negative  | -0.007 [-0.229, 0.216] |
| Dominant repository | Gate2 | Stirling largest repository | C/SE + PR     | 872 | Log_PR_Size  | AME    | -0.0351271  | 0.0992229  | 0.0289686  | 0.282759    | False       | negative  | -0.035 [-0.099, 0.029] |
| Dominant repository | Gate2 | Stirling top 5 repositories | C/SE + PR     | 692 | Context      | AME    | 0.107525    | -0.147341  | 0.362392   | 0.408301    | False       | positive  | 0.108 [-0.147, 0.362]  |
| Dominant repository | Gate2 | Stirling top 5 repositories | C/SE + PR     | 692 | Specificity  | AME    | 0.0379106   | 0.223147   | 0.298969   | 0.775932    | False       | positive  | 0.038 [-0.223, 0.299]  |
| Dominant repository | Gate2 | Stirling top 5 repositories | C/SE + PR     | 692 | Verification | AME    | -0.0558739  | 0.298017   | 0.186269   | 0.651084    | False       | negative  | -0.056 [-0.298, 0.186] |
| Dominant repository | Gate2 | Stirling top 5 repositories | C/SE + PR     | 692 | Log_PR_Size  | AME    | -0.0397048  | 0.109188   | 0.0297788  | 0.262724    | False       | negative  | -0.040 [-0.109, 0.029] |

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|----------------------|-------|---------------------------|----------------|-----|--------------|--------|-----------|-----------|----------|-------------|-------------|-----------|-----------------------|
| Language sensitivity | Gate0 | Exclude dominant language | LanguagePR 15e | 15e | Context      | OR     | 2.34888   | 1.22835   | 4.4916   | 0.009829    | True        | positive  | 2.35** [1.23, 4.49]   |
| Language sensitivity | Gate0 | Exclude dominant language | LanguagePR 15e | 15e | Specificity  | OR     | 5.36791   | 2.3497    | 12.2631  | 6.70061e-05 | True        | positive  | 5.37*** [2.35, 12.26] |
| Language sensitivity | Gate0 | Exclude dominant language | LanguagePR 15e | 15e | Verification | OR     | 0.737896  | 0.340224  | 1.60039  | 0.441604    | False       | negative  | 0.74 [0.34, 1.60]     |
| Language sensitivity | Gate0 | Exclude dominant language | LanguagePR 15e | 15e | Log_PR_SIR   | OR     | 0.994566  | 0.823759  | 1.20079  | 0.954805    | False       | negative  | 0.99 [0.82, 1.20]     |
| Language sensitivity | Gate0 | Exclude top 2 languages   | LanguagePR 12e | 12e | Context      | OR     | 2.26421   | 1.12098   | 4.57336  | 0.022705    | True        | positive  | 2.26* [1.12, 4.57]    |
| Language sensitivity | Gate0 | Exclude top 2 languages   | LanguagePR 12e | 12e | Specificity  | OR     | 4.74417   | 1.99418   | 11.2864  | 0.000430    | True        | positive  | 4.74*** [1.99, 11.29] |
| Language sensitivity | Gate0 | Exclude top 2 languages   | LanguagePR 12e | 12e | Verification | OR     | 0.763356  | 0.329736  | 1.76721  | 0.528376    | False       | negative  | 0.76 [0.33, 1.77]     |
| Language sensitivity | Gate0 | Exclude top 2 languages   | LanguagePR 12e | 12e | Log_PR_SIR   | OR     | 1.03484   | 0.844457  | 1.26814  | 0.741289    | False       | positive  | 1.03 [0.84, 1.27]     |
| Language sensitivity | Gate1 | Exclude dominant language | LanguagePR 10e | 10e | Context      | OR     | 2.44379   | 0.892526  | 6.69127  | 0.082083    | False       | positive  | 2.44 [0.89, 6.69]     |
| Language sensitivity | Gate1 | Exclude dominant language | LanguagePR 10e | 10e | Specificity  | OR     | 5.52459   | 1.49053   | 20.4766  | 0.010555    | True        | positive  | 5.52* [1.49, 20.48]   |
| Language sensitivity | Gate1 | Exclude dominant language | LanguagePR 10e | 10e | Verification | OR     | 4.66235   | 1.57444   | 13.8065  | 0.005445    | True        | positive  | 4.66** [1.57, 13.81]  |
| Language sensitivity | Gate1 | Exclude dominant language | LanguagePR 10e | 10e | Log_PR_SIR   | OR     | 1.37392   | 1.06138   | 1.7785   | 0.015851    | True        | positive  | 1.37* [1.06, 1.78]    |
| Language sensitivity | Gate1 | Exclude top 2 languages   | LanguagePR 7e  | 7e  | Context      | OR     | 2.36821   | 0.699347  | 8.01954  | 0.16595     | False       | positive  | 2.37 [0.70, 8.02]     |
| Language sensitivity | Gate1 | Exclude top 2 languages   | LanguagePR 7e  | 7e  | Specificity  | OR     | 8.84939   | 1.81052   | 43.2538  | 0.007076    | True        | positive  | 8.85** [1.81, 43.25]  |
| Language sensitivity | Gate1 | Exclude top 2 languages   | LanguagePR 7e  | 7e  | Verification | OR     | 4.10715   | 1.08907   | 15.4891  | 0.036983    | True        | positive  | 4.11* [1.09, 15.49]   |
| Language sensitivity | Gate1 | Exclude top 2 languages   | LanguagePR 7e  | 7e  | Log_PR_SIR   | OR     | 1.42316   | 1.04131   | 1.94504  | 0.026832    | True        | positive  | 1.42* [1.04, 1.95]    |
| Language sensitivity | Gate2 | Exclude dominant language | LanguagePR 5e  | 5e  | Context      | AME    | 0.145231  | -0.170196 | 0.460659 | 0.366834    | False       | positive  | 0.145 [-0.170, 0.46]  |
| Language sensitivity | Gate2 | Exclude dominant language | LanguagePR 5e  | 5e  | Specificity  | AME    | 0.036137  | 0.272994  | 0.345269 | 0.818776    | False       | positive  | 0.036 [-0.273, 0.34]  |
| Language sensitivity | Gate2 | Exclude dominant language | LanguagePR 5e  | 5e  | Verification | AME    | -0.043327 | 0.31138   | 0.224725 | 0.751391    | False       | negative  | -0.043 [-0.311, 0.22] |
| Language sensitivity | Gate2 | Exclude dominant language | LanguagePR 5e  | 5e  | Log_PR_SIR   | AME    | -0.032454 | 0.107507  | 0.042597 | 0.396694    | False       | negative  | -0.032 [-0.108, 0.04] |
| Language sensitivity | Gate2 | Exclude top 2 languages   | LanguagePR 4e  | 4e  | Context      | AME    | 0.104581  | -0.251062 | 0.460223 | 0.56438     | False       | positive  | 0.105 [-0.251, 0.46]  |
| Language sensitivity | Gate2 | Exclude top 2 languages   | LanguagePR 4e  | 4e  | Specificity  | AME    | -0.002504 | 0.358013  | 0.353003 | 0.988982    | False       | negative  | -0.003 [-0.358, 0.35] |
| Language sensitivity | Gate2 | Exclude top 2 languages   | LanguagePR 4e  | 4e  | Verification | AME    | -0.063381 | 0.369582  | 0.242819 | 0.684963    | False       | negative  | -0.063 [-0.370, 0.24] |
| Language sensitivity | Gate2 | Exclude top 2 languages   | LanguagePR 4e  | 4e  | Log_PR_SIR   | AME    | -0.016408 | 0.102114  | 0.069323 | 0.707563    | False       | negative  | -0.016 [-0.102, 0.07] |
| PR-size outlier      | Gate1 | Exclude top 1% PR-size    | PR-sizePR 12e  | 12e | Context      | OR     | 1.88154   | 0.781278  | 4.53129  | 0.158672    | False       | positive  | 1.88 [0.78, 4.53]     |
| PR-size outlier      | Gate1 | Exclude top 1% PR-size    | PR-sizePR 12e  | 12e | Specificity  | OR     | 7.29494   | 2.28295   | 23.3103  | 0.000800    | True        | positive  | 7.29*** [2.28, 23.31] |
| PR-size outlier      | Gate1 | Exclude top 1% PR-size    | PR-sizePR 12e  | 12e | Verification | OR     | 3.4223    | 1.33709   | 8.75943  | 0.010294    | True        | positive  | 3.42* [1.34, 8.76]    |
| PR-size outlier      | Gate1 | Exclude top 1% PR-size    | PR-sizePR 12e  | 12e | Log_PR_SIR   | OR     | 1.53098   | 1.18977   | 1.97005  | 0.000930    | True        | positive  | 1.53*** [1.19, 1.97]  |
| PR-size outlier      | Gate1 | Exclude top 5% PR-size    | PR-sizePR 12e  | 12e | Context      | OR     | 2.05501   | 0.840409  | 5.02503  | 0.11437     | False       | positive  | 2.06 [0.84, 5.03]     |
| PR-size outlier      | Gate1 | Exclude top 5% PR-size    | PR-sizePR 12e  | 12e | Specificity  | OR     | 7.41591   | 2.2919    | 23.9957  | 0.000824    | True        | positive  | 7.42*** [2.29, 24.00] |
| PR-size outlier      | Gate1 | Exclude top 5% PR-size    | PR-sizePR 12e  | 12e | Verification | OR     | 3.29565   | 1.27144   | 8.54251  | 0.014121    | True        | positive  | 3.30* [1.27, 8.54]    |
| PR-size outlier      | Gate1 | Exclude top 5% PR-size    | PR-sizePR 12e  | 12e | Log_PR_SIR   | OR     | 1.54184   | 1.18199   | 2.01124  | 0.001408    | True        | positive  | 1.54** [1.18, 2.01]   |
| PR-size outlier      | Gate2 | Exclude top 1% PR-size    | PR-sizePR 8e   | 8e  | Context      | AME    | 0.141749  | -0.095989 | 0.237948 | 0.242562    | False       | positive  | 0.142 [-0.096, 0.37]  |
| PR-size outlier      | Gate2 | Exclude top 1% PR-size    | PR-sizePR 8e   | 8e  | Specificity  | AME    | -0.027540 | 0.263924  | 0.208843 | 0.819372    | False       | negative  | -0.028 [-0.264, 0.20] |
| PR-size outlier      | Gate2 | Exclude top 1% PR-size    | PR-sizePR 8e   | 8e  | Verification | AME    | -0.000131 | 0.051127  | 0.212437 | 0.99903     | False       | negative  | -0.000 [-0.213, 0.21] |
| PR-size outlier      | Gate2 | Exclude top 1% PR-size    | PR-sizePR 8e   | 8e  | Log_PR_SIR   | AME    | -0.038843 | 0.101744  | 0.024055 | 0.226131    | False       | negative  | -0.039 [-0.102, 0.02] |
| PR-size outlier      | Gate2 | Exclude top 5% PR-size    | PR-sizePR 8e   | 8e  | Context      | AME    | 0.157059  | -0.089244 | 0.403359 | 0.211368    | False       | positive  | 0.157 [-0.089, 0.40]  |
| PR-size outlier      | Gate2 | Exclude top 5% PR-size    | PR-sizePR 8e   | 8e  | Specificity  | AME    | -0.027465 | 0.267596  | 0.212664 | 0.822619    | False       | negative  | -0.027 [-0.268, 0.21] |

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|--------------------|--------|------------------------|----------------------|-----|--------------|-------------|------------|-----------|-----------|-------------|-------------|--------------|------------------------|
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size | Size + PR-size       | 82e | Verification | AME         | 0.00877962 | 0.21609   | 0.233649  | 0.939003    | False       | positive     | 0.009 [-0.216, 0.233]  |
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size | Size + PR-size       | 82e | Log_PR_size  | AME         | -0.0399726 | 0.109758  | 0.0298122 | 0.261581    | False       | negative     | -0.040 [-0.110, 0.030] |
| PR-size outlier    | Gate 2 | Exclude top 1% PR-size | Size + PR-size       | 253 | Context      | HR          | 1.03554    | 0.815968  | 1.31419   | 0.773939    | False       | positive     | 1.04 [0.82, 1.31]      |
| PR-size outlier    | Gate 2 | Exclude top 1% PR-size | Size + PR-size       | 253 | Specificity  | HR          | 1.02405    | 0.775465  | 1.35233   | 0.866938    | False       | positive     | 1.02 [0.78, 1.35]      |
| PR-size outlier    | Gate 2 | Exclude top 1% PR-size | Size + PR-size       | 253 | Verification | HR          | 0.882244   | 0.666494  | 1.16783   | 0.381228    | False       | negative     | 0.88 [0.67, 1.17]      |
| PR-size outlier    | Gate 2 | Exclude top 1% PR-size | Size + PR-size       | 253 | Log_PR_size  | HR          | 0.933655   | 0.872344  | 0.999274  | 0.0475966   | True        | negative     | 0.93* [0.87, 1.00]     |
| PR-size outlier    | Gate 2 | Exclude top 1% PR-size | Size + PR-size       | 253 | Context      | HR          | 1.48284    | 0.800168  | 2.74796   | 0.210681    | False       | positive     | 1.48 [0.80, 2.75]      |
| PR-size outlier    | Gate 2 | Exclude top 1% PR-size | Size + PR-size       | 253 | Specificity  | HR          | 0.906063   | 0.481666  | 1.70439   | 0.759605    | False       | negative     | 0.91 [0.48, 1.70]      |
| PR-size outlier    | Gate 2 | Exclude top 1% PR-size | Size + PR-size       | 253 | Verification | HR          | 1.7671     | 0.989109  | 3.15704   | 0.0544784   | False       | positive     | 1.77 [0.99, 3.16]      |
| PR-size outlier    | Gate 2 | Exclude top 1% PR-size | Size + PR-size       | 253 | Log_PR_size  | HR          | 0.782506   | 0.678178  | 0.902883  | 0.0007812   | True        | negative     | 0.78*** [0.68, 0.90]   |
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size | Size + PR-size       | 230 | Context      | HR          | 1.0445     | 0.821248  | 1.32845   | 0.722663    | False       | positive     | 1.04 [0.82, 1.33]      |
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size | Size + PR-size       | 230 | Specificity  | HR          | 1.0379     | 0.785042  | 1.3722    | 0.793994    | False       | positive     | 1.04 [0.79, 1.37]      |
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size | Size + PR-size       | 230 | Verification | HR          | 0.848645   | 0.634111  | 1.13576   | 0.269681    | False       | negative     | 0.85 [0.63, 1.14]      |
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size | Size + PR-size       | 230 | Log_PR_size  | HR          | 0.949376   | 0.880759  | 1.02334   | 0.174692    | False       | negative     | 0.95 [0.88, 1.02]      |
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size | Size + PR-size       | 230 | Context      | HR          | 1.42539    | 0.748968  | 2.71273   | 0.280329    | False       | positive     | 1.43 [0.75, 2.71]      |
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size | Size + PR-size       | 230 | Specificity  | HR          | 0.926931   | 0.479887  | 1.79042   | 0.821277    | False       | negative     | 0.93 [0.48, 1.79]      |
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size | Size + PR-size       | 230 | Verification | HR          | 1.79405    | 0.987257  | 3.26017   | 0.0551216   | False       | positive     | 1.79 [0.99, 3.26]      |
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size | Size + PR-size       | 230 | Log_PR_size  | HR          | 0.741636   | 0.635524  | 0.865467  | 0.0001482   | True        | negative     | 0.74*** [0.64, 0.87]   |
| Gate 2 alternative | Gate 2 | Resizetories           | Ordinal logistic     | 37  | Context      | Coefficient | 0.969057   | 0.072384  | 1.86573   | 0.0341595   | True        | positive     | 0.97* [0.07, 1.87]     |
| Gate 2 alternative | Gate 2 | Resizetories           | Ordinal logistic     | 37  | Specificity  | Coefficient | 0.078072   | -0.888716 | 1.04486   | 0.87424     | False       | positive     | 0.08 [-0.89, 1.04]     |
| Gate 2 alternative | Gate 2 | Resizetories           | Ordinal logistic     | 37  | Verification | Coefficient | 0.268851   | -1.10083  | 0.563122  | 0.526498    | False       | negative     | -0.27 [-1.10, 0.56]    |
| Gate 2 alternative | Gate 2 | Resizetories           | Ordinal logistic     | 37  | Log_PR_size  | Coefficient | 0.240643   | -0.490898 | 0.0096129 | 0.0594735   | False       | negative     | -0.24 [-0.49, 0.01]    |
| Gate2 binary       | Gate 2 | Low vs high reuse      | Binary logistic      | 58  | Context      | OR          | 2.97465    | 0.85345   | 10.368    | 0.087042    | False       | positive     | 2.97 [0.85, 10.37]     |
| Gate2 binary       | Gate 2 | Low vs high reuse      | Binary logistic      | 58  | Specificity  | OR          | 1.06745    | 0.286793  | 3.97306   | 0.922457    | False       | positive     | 1.07 [0.29, 3.97]      |
| Gate2 binary       | Gate 2 | Low vs high reuse      | Binary logistic      | 58  | Verification | OR          | 0.934886   | 0.273385  | 3.19701   | 0.914528    | False       | negative     | 0.93 [0.27, 3.20]      |
| Gate2 binary       | Gate 2 | Low vs high reuse      | Binary logistic      | 58  | Log_PR_size  | OR          | 0.733346   | 0.50415   | 1.06674   | 0.104789    | False       | negative     | 0.73 [0.50, 1.07]      |
| Aggregate PQS      | Gate 0 | PQS only               | PQS + PR-size        | 214 | PQS          | OR          | 2.57452    | 1.91963   | 3.45282   | 2.71217e-10 | True        | positive     | 2.57*** [1.92, 3.45]   |
| Aggregate PQS      | Gate 0 | PQS only               | PQS + PR-size        | 214 | Log_PR_size  | OR          | 1.00717    | 0.846965  | 1.19768   | 0.935568    | False       | positive     | 1.01 [0.85, 1.20]      |
| Aggregate PQS      | Gate 0 | PQS only               | PQS + PR-size        | 129 | PQS          | OR          | 3.67143    | 2.33274   | 5.77836   | 1.90459e-08 | True        | positive     | 3.67*** [2.33, 5.78]   |
| Aggregate PQS      | Gate 0 | PQS only               | PQS + PR-size        | 129 | Log_PR_size  | OR          | 1.43602    | 1.13708   | 1.81356   | 0.0023764   | True        | positive     | 1.44** [1.14, 1.81]    |
| Aggregate PQS      | Gate 0 | PQS only               | PQS + PR-size        | 87  | PQS          | AME         | 0.0173553  | 0.078067  | 0.112778  | 0.721485    | False       | positive     | 0.017 [-0.078, 0.113]  |
| Aggregate PQS      | Gate 0 | PQS only               | PQS + PR-size        | 87  | Log_PR_size  | AME         | -0.0374837 | 0.097411  | 0.0224444 | 0.220231    | False       | negative     | -0.037 [-0.097, 0.022] |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) | Size + prompt length | 214 | Context      | DR          | 3.08965    | 1.71602   | 5.56281   | 0.000170035 | True        | positive     | 3.09*** [1.72, 5.56]   |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) | Size + prompt length | 214 | Specificity  | DR          | 5.86041    | 2.71352   | 12.6568   | 6.76341e-06 | True        | positive     | 5.86*** [2.71, 12.66]  |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) | Size + prompt length | 214 | Verification | DR          | 0.938345   | 0.454836  | 1.93585   | 0.863256    | False       | negative     | 0.94 [0.45, 1.94]      |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) | Size + prompt length | 214 | Log_PR_size  | DR          | 1.03165    | 0.862706  | 1.23368   | 0.732738    | False       | positive     | 1.03 [0.86, 1.23]      |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) | Size + prompt length | 129 | Context      | DR          | 1.09088    | 0.167952  | 7.0854    | 0.927402    | False       | positive     | 1.09 [0.17, 7.09]      |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) | Size + prompt length | 129 | Specificity  | DR          | 2.10493    | 0.834721  | 5.30806   | 0.114761    | False       | positive     | 2.10 [0.83, 5.31]      |

| Family             | Gate    | Scenario                     | Specification               | N    | Term          | Metric      | Estimate   | CI_Low    | CI_High   | p_value     | Significant | Direction | Formatted               |
|--------------------|---------|------------------------------|-----------------------------|------|---------------|-------------|------------|-----------|-----------|-------------|-------------|-----------|-------------------------|
| Prompt-length      | Control | Add log(prompt_tokens/V + PR | Size + prompt_length        | 182  | Spreading     | OR          | 4.56167    | 1.36552   | 15.2388   | 0.0136557   | True        | positive  | 4.56* [1.37, 15.24]     |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR | Size + prompt_length        | 182  | Verification  | OR          | 2.20911    | 0.797672  | 6.11801   | 0.127257    | False       | positive  | 2.21 [0.80, 6.12]       |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR | Size + prompt_length        | 182  | prompt_length | SE          | 1.4702     | 1.14507   | 1.88765   | 0.00250897  | True        | positive  | 1.47** [1.15, 1.89]     |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR | Size + prompt_length        | 182  | prompt_length | Tokens      | 86.9035    | 5.29031   | 1427.56   | 0.00176893  | True        | positive  | 86.90** [5.29, 1427.56] |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR | Size + prompt_length        | 87   | Context       | AME         | 0.137797   | -0.096295 | 0.37189   | 0.248614    | False       | positive  | 0.138 [-0.096, 0.37]    |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR | Size + prompt_length        | 87   | Spreading     | AME         | -0.0138204 | 0.258945  | 0.231304  | 0.912009    | False       | negative  | -0.014 [-0.259, 0.231]  |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR | Size + prompt_length        | 87   | Verification  | AME         | 0.00387350 | 0.208835  | 0.216582  | 0.971528    | False       | positive  | 0.004 [-0.209, 0.216]   |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR | Size + prompt_length        | 87   | prompt_length | Size        | -0.0366493 | 0.0980523 | 0.0247544 | 0.242071    | False       | negative  | -0.037 [-0.098, 0.024]  |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR | Size + prompt_length        | 87   | prompt_length | Tokens      | -0.109059  | 0.624108  | 0.40599   | 0.678132    | False       | negative  | -0.109 [-0.624, 0.406]  |
| Hold-out stability | Gate 0  | Stratified 80/20 split       | Train N=1711, Holdout N=413 | 1711 | Context       | Coefficient | 1.16641    | nan       | nan       | 0.00106235  | True        | positive  | beta=1.164**            |
| Hold-out stability | Gate 0  | Stratified 80/20 split       | Train N=1711, Holdout N=413 | 1711 | Specificity   | Coefficient | 1.78173    | nan       | nan       | 5.57349e-05 | True        | positive  | beta=1.782***           |
| Hold-out stability | Gate 0  | Stratified 80/20 split       | Train N=1711, Holdout N=413 | 1711 | Verification  | Coefficient | 0.0499011  | nan       | nan       | 0.914272    | False       | positive  | beta=0.044              |
| Hold-out stability | Gate 0  | Stratified 80/20 split       | Train N=1711, Holdout N=413 | 1711 | Log PR        | Coefficient | 0.100722   | nan       | nan       | 0.352096    | False       | negative  | beta=-0.101             |
| Hold-out stability | Gate 1  | Stratified 80/20 split       | Train N=1121, Holdout N=277 | 1121 | Context       | Coefficient | 0.461963   | nan       | nan       | 0.347963    | False       | positive  | beta=0.462              |
| Hold-out stability | Gate 1  | Stratified 80/20 split       | Train N=1121, Holdout N=277 | 1121 | Specificity   | Coefficient | 2.39983    | nan       | nan       | 0.00210017  | True        | positive  | beta=2.390**            |
| Hold-out stability | Gate 1  | Stratified 80/20 split       | Train N=1121, Holdout N=277 | 1121 | Verification  | Coefficient | 1.38551    | nan       | nan       | 0.00897343  | True        | positive  | beta=1.386**            |
| Hold-out stability | Gate 1  | Stratified 80/20 split       | Train N=1121, Holdout N=277 | 1121 | Log PR        | Coefficient | 0.316369   | nan       | nan       | 0.0170656   | True        | positive  | beta=0.316*             |
| Hold-out stability | Gate 2  | Stratified 80/20 split       | Train N=69, Holdout N=15    | 69   | Context       | Coefficient | 0.572621   | nan       | nan       | 0.378436    | False       | positive  | beta=0.573              |
| Hold-out stability | Gate 2  | Stratified 80/20 split       | Train N=69, Holdout N=15    | 69   | Specificity   | Coefficient | 0.211712   | nan       | nan       | 0.726954    | False       | negative  | beta=-0.212             |
| Hold-out stability | Gate 2  | Stratified 80/20 split       | Train N=69, Holdout N=15    | 69   | Verification  | Coefficient | 0.0860099  | nan       | nan       | 0.888448    | False       | positive  | beta=0.081              |
| Hold-out stability | Gate 2  | Stratified 80/20 split       | Train N=69, Holdout N=15    | 69   | Log PR        | Coefficient | 0.102465   | nan       | nan       | 0.237444    | False       | negative  | beta=-0.192             |